

Methodological Notes for Event Tests on Daily CARDI

CARDI Project

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1 Purpose

This note documents the econometric methods implemented in `Code/Plot_Code/Event_test.R`. The goal is to test whether carbon-related regulatory events are associated with structural changes in the daily CARDI time series. The script uses three daily CARDI series:

$$\text{CARDI}_{1P,t}, \quad \text{CARDI}_{5P,t}, \quad \text{CARDI}_{10P,t}.$$

The script implements four empirical components:

1. event clustering and overlap diagnostics;
2. known-breakpoint Chow tests;
3. interrupted time-series regressions with Newey–West standard errors;
4. two-regime Markov-switching regime analysis.

It also preserves a reference-style event-dummy regression based on smoothed CARDI log changes, with the important modification that event-dummy coefficients are reported using Newey–West standard errors.

2 Data

Let $y_t^{(q)}$ denote the CARDI series for quantile level $q \in \{1P, 5P, 10P\}$. The daily CARDI data are loaded from:

`Data/Processed/FRM_Carbon_risk.csv`.

The event data are loaded from:

`Data/raw/Important_Carbon_Events.xlsx`.

Let $\mathcal{E} = \{e_1, \dots, e_J\}$ denote the set of event dates. If an event occurs on a non-trading day, the script maps the event to the nearest observed CARDI trading date.

3 Event Clustering

Several policy events occur close to one another. Treating nearby events as independent shocks can make local event-window tests difficult to interpret. The script therefore constructs event clusters before running the main tests.

Let the sorted event dates be:

$$e_1 \leq e_2 \leq \dots \leq e_J.$$

For a configurable clustering threshold G , currently set to 30 calendar days, event e_j belongs to the same cluster as e_{j-1} if

$$e_j - e_{j-1} \leq G.$$

Otherwise, e_j starts a new cluster.

For each cluster c , the script records:

$$(\text{cluster_id}_c, \text{start}_c, \text{end}_c, \text{representative_date}_c, N_c),$$

where N_c is the number of events in the cluster. The representative event date is the first event date in the cluster.

4 Known-Breakpoint Chow Test

4.1 Motivation

The Chow test evaluates whether regression coefficients are stable across a known breakpoint. The test was introduced by Chow (1960). In this application, each event date is treated as a known candidate breakpoint.

4.2 Local Event Window

For an event date e and a symmetric trading-day window h , the local sample is:

$$\mathcal{T}(e, h) = \{t : -h \leq \tau_t(e) \leq h\},$$

where $\tau_t(e)$ is the trading-day distance between date t and the nearest CARDI trading date for event e .

4.3 Restricted and Unrestricted Models

For each CARDI variable $y_t^{(q)}$, the restricted model is:

$$y_t^{(q)} = \alpha + \beta t + u_t. \tag{1}$$

The unrestricted model allows both the intercept and slope to change after the event:

$$y_t^{(q)} = \alpha + \beta t + \gamma D_t(e) + \delta [t \times D_t(e)] + u_t, \tag{2}$$

where:

$$D_t(e) = \begin{cases} 1, & t \geq e, \\ 0, & t < e. \end{cases}$$

4.4 Test Statistic

Let RSS_R be the residual sum of squares from the restricted model and RSS_U be the residual sum of squares from the unrestricted model. The Chow statistic is:

$$F = \frac{(RSS_R - RSS_U)/q}{RSS_U/(n - k_U)}, \quad (3)$$

where q is the number of restrictions, n is the local sample size, and k_U is the number of parameters in the unrestricted model.

The null hypothesis is:

$$H_0 : \gamma = 0 \quad \text{and} \quad \delta = 0.$$

Rejecting H_0 indicates evidence of a known structural break around the event date.

5 Interrupted Time-Series Regressions

5.1 Motivation

Interrupted time-series models are commonly used to estimate whether an intervention is associated with an immediate change, a persistent level shift, or a trend change. See Bernal, Cummins, and Gasparrini (2017) for a tutorial treatment of segmented regression designs for intervention analysis.

5.2 Regression Specification

For each CARDI variable $y_t^{(q)}$, event e , and local window $\mathcal{T}(e, h)$, the script estimates:

$$y_t^{(q)} = \alpha + \beta_1 \text{Pulse}_{t,e} + \beta_2 \text{Post}_{t,e} + \beta_3 \text{Trend}_{t,e} + \varepsilon_t. \quad (4)$$

The event variables are defined as:

$$\text{Pulse}_{t,e} = \mathbf{1}\{t = e\}, \quad (5)$$

$$\text{Post}_{t,e} = \mathbf{1}\{t > e\}, \quad (6)$$

$$\text{Trend}_{t,e} = \max\{\tau_t(e), 0\}. \quad (7)$$

The coefficients have the following interpretation:

- β_1 : same-day event shock;
- β_2 : persistent post-event level shift;
- β_3 : post-event trend change.

5.3 Newey–West Standard Errors

Because daily CARDI is a time series, residuals may be serially correlated and heteroskedastic. The script therefore reports Newey–West heteroskedasticity and autocorrelation consistent standard errors (Newey and West, 1987).

Let X be the matrix of regressors and $\hat{\varepsilon}_t$ be the OLS residuals. The Newey–West covariance estimator can be written as:

$$\widehat{\text{Var}}_{\text{NW}}(\hat{\beta}) = (X'X)^{-1} \left(\sum_{\ell=-L}^L w_\ell \sum_{t=|\ell|+1}^T x_t x'_{t-|\ell|} \hat{\varepsilon}_t \hat{\varepsilon}_{t-|\ell|} \right) (X'X)^{-1}, \quad (8)$$

where L is the truncation lag and the Bartlett weights are:

$$w_\ell = 1 - \frac{|\ell|}{L+1}. \quad (9)$$

The script uses:

$$L = 5.$$

6 Reference-Style Event-Dummy Regression

6.1 Smoothed CARDI Series

The script preserves the legacy plotting and regression logic from the earlier `Event_test.R`. First, each CARDI series is smoothed using LOESS over the daily time index:

$$\tilde{y}_t^{(q)} = \text{LOESS} \left(y_t^{(q)}; \text{span} = 0.1 \right). \quad (10)$$

The dependent variable is the log change in smoothed CARDI:

$$\Delta \tilde{y}_t^{(q)} = 100 \times \left[\log \left(\tilde{y}_t^{(q)} \right) - \log \left(\tilde{y}_{t-1}^{(q)} \right) \right]. \quad (11)$$

6.2 Trailing Event Dummy

For each lookback window $m \in \{7, 30, 60\}$, define:

$$\text{DummyEvent}_{t,m} = \mathbf{1} \{ \exists e_j \in \mathcal{E} \text{ such that } t - (m - 1) \leq e_j \leq t \}. \quad (12)$$

The regression is:

$$\Delta \tilde{y}_t^{(q)} = \alpha + \theta_m \text{DummyEvent}_{t,m} + \eta_t. \quad (13)$$

The coefficient θ_m measures whether smoothed CARDI log changes are different on days following recent carbon-related events. The script reports Newey–West standard errors for θ_m .

7 Markov-Switching Regime Analysis

7.1 Motivation

The Markov-switching model tests whether CARDI alternates between low-risk and high-risk latent regimes. The approach follows the regime-switching framework introduced by Hamilton (1989).

7.2 Two-Regime Model

For each CARDI series, the script estimates a two-regime model:

$$y_t^{(q)} = \mu_{s_t}^{(q)} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_{s_t}^2), \quad (14)$$

where $s_t \in \{1, 2\}$ is an unobserved Markov state.

The state evolves according to:

$$\Pr(s_t = j \mid s_{t-1} = i) = p_{ij}, \quad i, j \in \{1, 2\}. \quad (15)$$

The transition matrix is:

$$P = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}. \quad (16)$$

7.3 High-Risk Regime Identification

After estimation, the script extracts smoothed state probabilities:

$$\Pr(s_t = j \mid \mathcal{F}_T),$$

where $\mathcal{F}_T$ denotes the full sample information set.

The high-risk regime is defined as the regime with the higher probability-weighted CARDI mean:

$$j^* = \arg \max_{j \in \{1,2\}} \frac{\sum_t \Pr(s_t = j \mid \mathcal{F}_T) y_t}{\sum_t \Pr(s_t = j \mid \mathcal{F}_T)}. \quad (17)$$

The high-risk probability is:

$$\pi_t^H = \Pr(s_t = j^* \mid \mathcal{F}_T). \quad (18)$$

7.4 Regime Switch Dates

For a threshold $c \in \{0.5, 0.8\}$, a switch into the high-risk regime is defined by:

$$\pi_t^H \geq c \quad \text{and} \quad \pi_{t-1}^H < c. \quad (19)$$

Each switch date is matched to the nearest individual event and nearest event cluster. The script reports both calendar-day and trading-day distances.

8 Output Files

Output	Description
<code>event_clusters.xlsx</code>	Individual and clustered event metadata
<code>chow_known_breakpoint_tests.xlsx</code>	Known-breakpoint Chow test results
<code>intervention_newey_west_tests.xlsx</code>	Interrupted time-series regressions
<code>reference_event_dummy_newey_west.xlsx</code>	Legacy event-dummy regressions with HAC SEs
<code>markov_switching_results.xlsx</code>	Regime probabilities and switch-event matches
<code>FRM_Event.csv</code>	Smoothed CARDI series and log differences
<code>RegEvent_SmoothFRMDiff*.csv</code>	Reference event-dummy regression outputs
<code>Plot_Event.png</code>	Legacy CARDI event plot
<code>MarkovSwitching_CARDI_*.png</code>	Markov regime probability plots

9 Interpretation

The tests answer related but distinct questions. The Chow test asks whether the local regression relationship changes around a known event date. The interrupted time-series regressions decompose the event association into an immediate pulse, a persistent level shift, and a post-event trend change. The reference event-dummy model asks whether recent event occurrence predicts changes in smoothed CARDI. The Markov-switching model asks whether CARDI enters a high-risk latent regime and whether such regime transitions occur near known events.

References

References

- Bernal, J. L., Cummins, S., and Gasparrini, A. (2017). Interrupted time series regression for the evaluation of public health interventions: a tutorial. *International Journal of Epidemiology*, 46(1), 348–355. <https://doi.org/10.1093/ije/dyw098>
- Chow, G. C. (1960). Tests of equality between sets of coefficients in two linear regressions. *Econometrica*, 28(3), 591–605. <https://doi.org/10.2307/1910133>
- Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357–384. <https://doi.org/10.2307/1912559>
- Newey, W. K., and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708. <https://doi.org/10.2307/1913610>